

Theorem (Divergence/Gauss theorem). For a smooth vector field $\mathbf{F}(\mathbf{r})$,

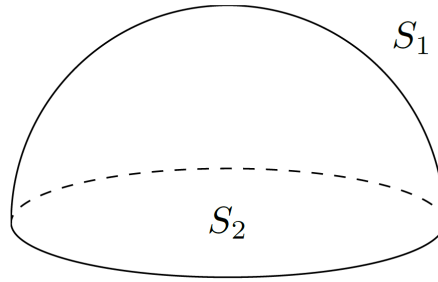
$$\int_V \nabla \cdot \mathbf{F} dV = \int_{\partial V} \mathbf{F} \cdot \mathbf{n} dS,$$

where V is a bounded volume with boundary ∂V , a piecewise smooth, closed surface, with outward normal $\mathbf{n}$.

Example. Consider the vector field $\mathbf{F} = (0, 0, z + a)$. The volume V is a solid hemisphere

$$x^2 + y^2 + z^2 \leq a^2, \quad z \geq 0,$$

with boundary $\partial V = S_1 \cup S_2$, the top curved surface of the hemisphere and the disc at the bottom.



The divergence is equal to $\nabla \cdot \mathbf{F} = 1$. The volume integral,

$$\int_V \nabla \cdot \mathbf{F} dV = \boxed{\frac{2}{3}\pi a^3},$$

is equal to the volume of the hemisphere.

On the curved surface S_1 ,

$$\mathbf{n} dS = \frac{1}{a}(x, y, z) dS.$$

$$\Rightarrow \mathbf{F} \cdot \mathbf{n} dS = \frac{1}{a}z(z + a) dS = a \cos \theta (\cos \theta + 1) \underbrace{a^2 \sin \theta d\theta d\varphi}_{dS}.$$

$$\begin{aligned} \Rightarrow \int_{S_1} \mathbf{F} \cdot \mathbf{n} dS &= a^3 \int_0^{2\pi} d\varphi \int_0^{\pi/2} \sin \theta (\cos^2 \theta + \cos \theta) d\theta \\ &= 2\pi a^3 \left[\frac{-1}{3} \cos^3 \theta - \frac{1}{2} \cos^2 \theta \right]_0^{\pi/2} = \boxed{\frac{5}{3}\pi a^3} \end{aligned}$$

On the flat disc S_2 ,

$$\mathbf{n} dS = -(0, 0, 1) dS.$$

Then, on $z = 0$, we have

$$\mathbf{F} \cdot \mathbf{n} dS = -a dS.$$

Hence,

$$\int_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \boxed{-\pi a^3}.$$

Finally, check that

$$\int_{S_1} \mathbf{F} \cdot \mathbf{n} dS + \int_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \boxed{\frac{5}{3}\pi a^3} + \boxed{-\pi a^3} = \boxed{\frac{2}{3}\pi a^3},$$

in accordance with the Divergence Theorem.